CSC362 Final Project by Asuna Dai and Umurerwa Kalisa Tania

Final Visualization: Video Game Sales By Genre and Region

- Link to Asuna's viz page: https://tianyedai.github.io/
- Link to Tania's viz page: taumurerwakalisa.github.io

Finding The Dataset

To begin with, our pair did not have any initial ideas on what we wanted to visualize or how we wanted to visualize it. So we started off with searching the web for datasets.
Finally, our search led us to a dataset of video game sales that was published on the website called zenodo.org in January 2022, and was put together by Gregory Smith.

The Dataset

The 'Video Game Sales' dataset contained a list of video games whose sales were greater than 100,000 copies. The dataset was compiled by scraping data from vgchartz.com and had 16,598 records/items.

The fields in this dataset include:

- Rank: Ranking of overall sales. Video games were ranked based on the overall (global sales) from the highest selling game to the lowest selling game.
- Name: The name of the game.
- Platform: The Platform on which the game was released (i.e. PC,PS4, etc.)
- Year: Year of the game's release.
- Genre: Genre of the game.
- Publisher: Publisher of the game.
- NA_Sales: Sales made in North America (in millions).
- EU_Sales - Sales made in Europe (in millions).
- JP_Sales - Sales made in Japan (in millions).
- Other_Sales - Sales made in the rest of the world (in millions).
- Global_Sales - Total worldwide sales.

Cleaning the Dataset

Our dataset had a total of 16,598 items. After downloading the csv file and exploring the file in Excel, we noticed 197 items that had missing data for either the 'Year', 'EU_Sales', and 'NA_Sales'. Using the Excel 'Find' and 'Replace' functionality, we removed the records that had missing data. This brought our final item count to 16,401.
We also identified that there were twelve video game genres. This would be relevant for the development of our visualization.

Paper Prototypes

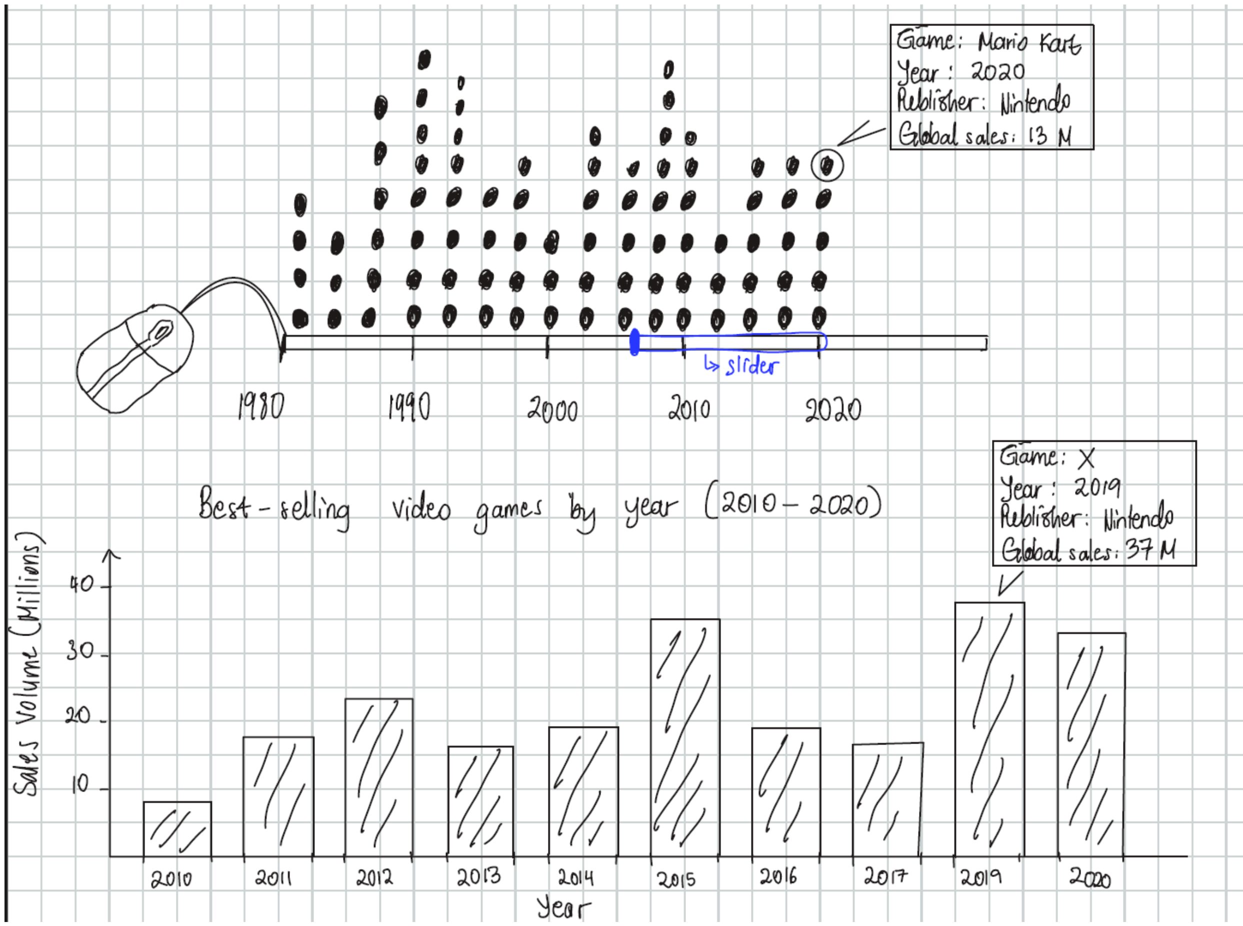
We developed two prototypes for our visualization: one we fully developed and one which we did not fully develop because we believed that it would not fit well enough with the structure of our dataset and intended tasks.
Below are descriptions of the two prototypes:

Prototype 1: A Timeline Visualization of Video Game Sales by Genre and Region

In this prototype, we suggested a timeline visualization. We took a more creative approach with this one. Below are some of the features of this prototype:

- A wired mouse as the interface for displaying the information about video games, based on the year.
- The wire part of the mouse would act as a mark that would be used to display the years within which a video game was made.
- Since we have video games from so many different years (1990-2020), we decided to bracket the years into decades. That is, 1980-1990, 1990-2000, 2000-2010, 2010-2020. The wire mouse would act as a channel for horizontal position (i.e. x-axis), displaying the four different year brackets.
- A slider would be added below the x-axis to allow a user to scope over different years and view the sales made in those years in much more detail.
- On clicking a specific time period within the slider, the user would activate a bar chart visualization that shows the best-selling video games in each specific year.

Fig.1. Prototype 1



Prototype 2: A Tree Map Visualization of Video Game Sales by Genre and Region

In this prototype, we suggested using a treemap to show the video game sale performance. This would be a great way to distinguish between video game sales because of the use of size to differentiate between higher and lower selling games. Below is a description of our prototype:

A. Default/Landing Page

The landing page has the title of our visualization, and a description of the visualization's purpose and how to navigate to the treemap. The main purpose of this page is to showcase the 12 game genres, and allow a user to select a single game and thereafter, transition to the treemap view. The treemap then displays the video game sales of that specific genre alone.

In this page, the genres are displayed as individual hexagons with the same size. The hexagons are arranged similar to a hex grid, but with separated hexagons. Along with the name of the genre, the number of games in that genre is displayed. Each hexagon has a different color based on a categorical color scheme.

B. **Treemap view**

This page is activated when a user has already selected a genre for which they wish to view video game sales. In the treemap, size/2D area encodes the number of sales a video game made in a specific region i.e. globally, Europe, North America, Japan, and the rest of the world (other). Color encodes the genre type i.e. the color of the area occupied by a video game in the treemap is inherited from the color of its genre (that is, the hexagon from default page). Additionally, color hue is used to encode the sales volume. That means video games with higher sales have darker hues while those with lower sales have lighter hues.

We also planned to add a tooltip feature to the treemap. So, when a user hovers over a videogame in the treemap, a tooltip appears showing information about the game's name, sale amount for a specific region, published year, platform, and publisher.

Our treemap also supports filter functions. There are three filter options:

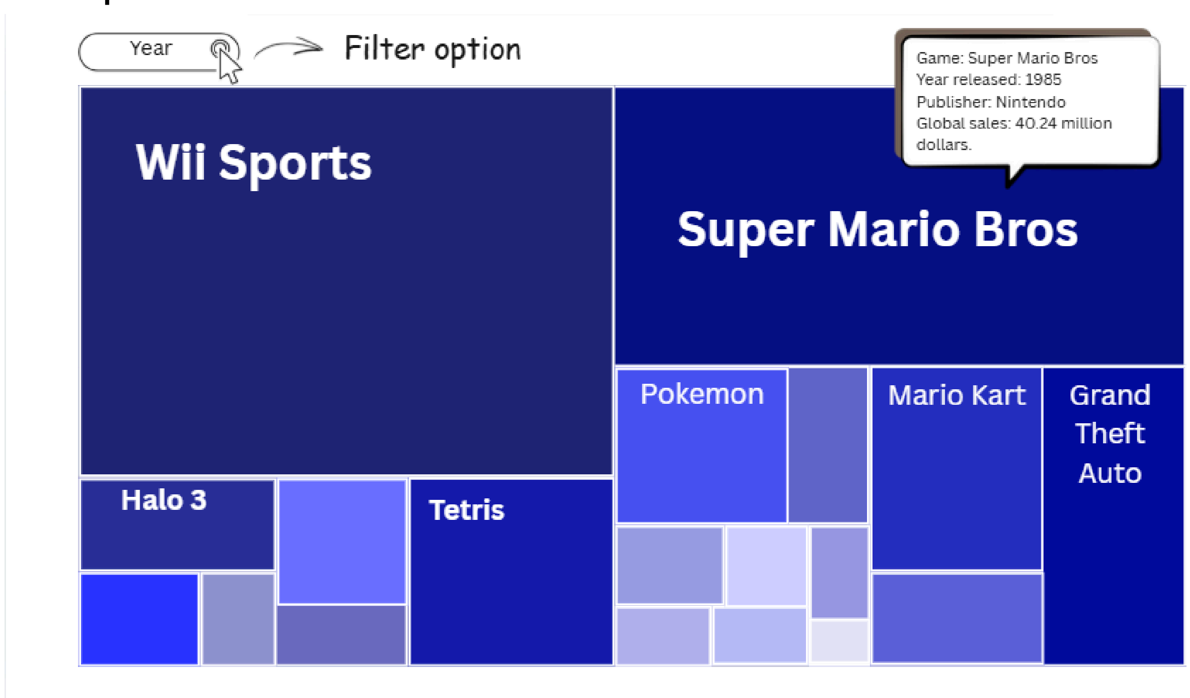
- Filter by sales region: A user can choose to view the sales made in either Europe, North America, Japan, other regions, or globally. The default view will display global sales.
- Filter by year: We will bracket the publishing years of games by decade i.e. 1980-1990, 1990-2000, 2000-2010, and 2010-2020 in order to allow a user to compare sales based on when a game was released. There will also be an option to filter by "All Years." The default view will display "All Years."

The treemap view also includes a legend that displays genre names and their corresponding colors.

After comparing the two prototype ideas, we chose to develop the 2nd prototype i.e. the Treemap. This was presented in our Presentation #4.

Fig.2: A screenshot of our treemap visualization prototype

Treemap View:



Task Analysis

For our Treemap visualization of video game sales by region and year, we identified the following tasks that our user should be able to accomplish:

1. Compare values: A user should be able to compare video game sales based on the size of the area occupied by a specific game.
2. Range: A user should be able to filter by year or region and identify the video game with the lowest and highest sales.
3. Order: A user should be able to infer the ranking of video games by sales volume (under some filtered attribute) by observing the size of the blocks and using the information on the tooltips in the treemap.

Accessibility Features

Our visualization will support the following accessibility features:

- Keyboard navigation: By pressing the tab, enter, and arrow buttons on the keyboard, users should be able to navigate through filter choices and blocks in the treemap, including the activation of the tooltips. The same keyboard buttons can be used to navigate the default page to allow a user to choose a video game genre.
- Screen reader accessibility: Our visualization will include appropriate alt and aria text labels and descriptions that can be read by a screen reader and guide users with visual impairments.
- Audio support: Incorporate a audio/non-audio mode in our visualization. If the audio mode is enabled, there will be an audio narrating the overview of the visualization and tips to navigate/interact with the visualization. The narrator can guide the user through keyboard navigation of the tree map and whenever a specific box is selected, the screen reader will read out loud the text in the blocks inside the treemap, as well as the text in the tooltip when prompted.

Implementation Process of The Visualization

Presentation #4

We presented our treemap prototype in class on April 13 and got the following feedback from some students and Dr. Williams:

- Consider creating a treemap that shows video games from all genres rather than a single genre. This will allow for a user to make sales comparisons across genres.
- Professor Williams informed us of the existing built in support for creating a treemap in D3.
- Delete the video game records that have missing data.

- Think about how to handle situations where video games are in more than one genre → Fortunately, we did not run into this issue because all video games were categorized under a single genre.
- In the default/landing page, consider having the hexagons in different sizes, based on how many games a genre has.

Improvements based on the feedback

After receiving feedback on our prototype, we decided to make the following changes to our suggested prototype, which we developed during the implementation process:

- Due to the vast number of video games in our dataset, we decided not to create a single treemap that showed video games across all genres. However, we made the following compromises:
    - ➔ In our landing page, a user will be able to choose multiple genres as opposed to only choosing one.
    - ➔ To account for the large number of video games and the additional option to choose multiple genres, we decided to only show either the top 10, 30, or 50 video games in the treemap. There will be a filter option to choose the number of games to view out of the three options mentioned above.

- In the default/landing page, the hexagons will have different sizes based on the number of video games in a genre.
- We removed all the video game records with missing data, which brought our dataset from 16,598 to 16,401 items.
- Since the visualization no longer requires only the selection of a single genre, this means that we need to add a way to navigate to the treemap view without the use of a transition. In light of this, we added the following functionalities:
    - ➔ Single clicking a genre to select it.
    - ➔ Double click a genre to view the corresponding treemaps.
    - ➔ If a user wants to view multiple genres e.g. 3 genres, they can single click each genre to select them individually then double click any of the three genres to navigate to the treemap.

Pilot Testing

We did our pilot testing on Monday, May 4 in class. There were 9 participants in total. For some participants, we went off script a little and allowed them to explore the visualization on their own or complete a slightly different task. Below is the script we used:

> **Speaker:** "We are evaluating our visualization and are asking you, the participant, to complete some tasks using the visualization and then

provide feedback about the visualization and experience. As a reminder, we are evaluating the visualization, not you as a participant, so you don't need to worry about being "right" as you complete these tasks. There are three tasks, followed by a brief feedback session. The whole pilot session should take under 5 minutes. Do you consent to participate?"

[Wait for yes]

Speaker: "Thank you for agreeing to participate. We will start with the three tasks. Please 'think aloud' as you complete the task, meaning voice what you are thinking as you work through the task.

1. Your first task is: Select one or more game genres of your choice in order to view the treemap showing its video game sales. Can you name at least 2 games in the tree map and identify their sales (e.g. global), the year they were published, and the name of the publisher?
    - Select two genres: For the top selling video games for each genre, compare their sales. Which one has more sales than the other?

    [Wait for user to end task]
2. Your second task is: Identify the video game with the largest and lowest global sales in the treemap. What were their respective sales?
    - Repeat the same process, but this time, identify the largest and lowest selling video game for the years 1990-2000.

[Wait for user to end task]

3. Your third task is: In this treemap, rank the top 10 video games based on highest global sales, then EU sales, and lastly, JP sales.
    - Repeat the same process, but this time, rank the top 5 video games that were made in 2010-2020.

[Wait for user to end task]

Speaker: "That is the end of the third task. For this last bit, we welcome any feedback you may have about the visualization or about your process for completing the tasks."

[Allow participant to speak first, then informal discussion]

- Their performance for the first task:
  The participants were mostly able to select one or more genres of their interest. They then double clicks to jump to the treemap page. However, some of them fail to do so at their first attempt to go to the treemap page because they will have to double click on one of the genre buttons, instead of a random place in the page. Some other participants jumped to the treemap page before they selected all genres they were interested in because they thought double click was for selecting the genres.
  Once they go to the treemap page, they are mostly able to figure out that hovering on the video game blocks gives details of the game, and is able to figure out sales of the games quickly. Some participants asked about the saturation of the color of the blocks. They were able to compare the amount of sales by observing the block sizes.

- Their performance for the second task:
  The participants were able to find the region filter and select “global sales.” They then find games with the largest and lowest global sales by observing the size of the block, and finding the blocks of the biggest size and smallest size in the treemap, respectively. Since games with lower sales had lower saturation and the video game name labels were in white, some participants said that it was a bit hard to read the labels.
- Their performance for the third task:

We did not ask them to rank the top 10 for all sale regions because that would take too much time in class, but we did ask them to do top 5 rankings on a specific time period and region, varied by each participant. They first applied the corresponding filter, and then figured out the exact rankings by observing the block size and, when block sizes are close, hover on the blocks to compare specific sales amounts.

The feedbacks we get from the participants:
- The genre options are a bit confusing to navigate. Instead of using double click to navigate to the treemap page, it might be better to have a “view treemap” button.
- In the treemap, some games look like they have similar sizings, even though they actually have different sales (This is mostly for games with lower sales).
- It seems like the variation of the saturation of colors to reflect sales amount is not necessary, because that is already encoded by the block size.
- Have bigger fonts for genres.
- Have bigger genre selection buttons.
- Provide more text instructions on how to use and navigate the viz.
- It might be better to place the treemap right under the genre selection buttons, instead of navigating to a separate page.
- Make the name labels of the blocks in the treemap bigger.
- The genre page seems a bit too empty.
- The color of the treemap is overall too pale.

List of changes we planned to make (and have already made):
- Initially, the genre page required double clicking to navigate to the treemap page. We added a “view X genres” button (where X is the number of genres selected) that the user can click after selecting the genres by single clicking.
- We made the genre page and the genre selection buttons larger. We have also adjusted the style so that the currently selected genres are more obviously shown.
- We deleted the color saturation encodings because it caused confusion.
- We added more text instructions on how to use and navigate the viz.
- We edit the legend so that when a user selects a genre, the legend on the treemap highlights the genres that were chosen, and the others are disabled.

Implementation of Accessibility Features

After the pilot testing phase, we implemented the following accessibility features:

1. **Keyboard navigation:** Both the default page and the treemap view can be navigated using keyboard buttons. Specifically,
    - On the default page that shows the genre names, a user can click the ‘Tab’ key to move around the page. By clicking enter, the user can select or deselect a hexagon in order to choose a genre. The user can also use the ‘Enter’ key to click the ‘view X genres’ button and view the treemap.
    - On the treemap view, clicking the ‘Tab’ key will allow a user to navigate through the page e.g. access filter options and legend. By clicking ‘Enter,’ the user can choose a specific option amongst the filter options. Additionally, the user can tab through the treemap to view individual blocks of video game sales; the block a user lands on will then be highlighted. The tooltip also appears when a game is highlighted.
2. **Screen reader accessibility:** Aria labels were added to parts of the visualization we thought were important for a user to know. To complement the aria labels, we added some aria-describedBy attributes to give more detailed descriptions about the visualization where necessary. We also added roles to various parts of the visualization. Some of the roles included region and tooltip.
3. **Zooming**: We adjusted the font size so that users can zoom to 400% and still accomplish the tasks.
4. **Descriptive Texts**: We have added texts explaining the purpose and functions of the visualization that is understandable for a broad audience. Instructions on ways to navigate the visualization are also provided.

Note: Due to the timeline we had to develop our visualization, we were not able to implement the audio support.

WAVE Testing

We used the WAVE extension to test our visualization. The WAVE did not detect any errors. It was able to identify all the aria attributes and roles in our website. We also passed the contrast test with a contrast of 8.59:1

User Testing

After making necessary improvements and adding accessibility features to our visualization, we conducted user testing with three different users who are not part of our CSC362 class.
We used the following script:

> **Speaker:** “We are evaluating our visualization and are asking you, the participant, to complete some tasks using the visualization and then provide feedback about the visualization and experience. As a reminder, we are evaluating the visualization, not you as a participant, so you don’t need to worry about being “right” as you complete these tasks. There are three tasks, followed by a brief feedback session. The whole

pilot session should take under 5 minutes. Do you consent to participate?”

[Wait for yes]

Speaker: “Thank you for agreeing to participate. We will start by giving you an overview of the visualization. Following that, you will be asked to complete four tasks. Please ‘think aloud’ as you complete a task, meaning voice what you are thinking as you work through the task.

1. Your first task is: Select one or more game genres of your choice in order to view the treemap showing its video game sales. Can you name at least 2 games in the tree map and identify their sales (e.g. global), the year they were published, and the name of the publisher?
   - Select two genres: For the top selling video games for each genre, compare their sales. Which one has more sales than the other?

   [Wait for user to end task]
2. Your second task is: Identify the video game with the largest and lowest global sales in the treemap. What were their respective sales?
   - Repeat the same process, but this time, identify the largest and lowest selling video game for the years 1990-2000.

[Wait for user to end task]

3. Your third task is: In this treemap, rank the top 10 video games based on highest global sales, then EU sales, and lastly, JP sales.
   - Repeat the same process, but this time, rank the top 5 video games that were made in 2010-2020.

[Wait for user to end task]

4. Your fourth task is: Use the keyboard to navigate the visualization. Use the ‘tab’ key to move around the page. Use the ‘enter’ or ‘select’ key to activate a button or choose a filter option for the treemap.

[Wait for user to end task]

Speaker: “That is the end of the fourth task. For this last bit, we welcome any feedback you may have about the visualization or about your process for completing the tasks.”

User 1:

- *Major/minor:* Biology major, Public Health minor (Pre-med track).
- *Notes on task completion:* This user was able to complete all tasks successfully. However, when they reached the treemap view, they pointed out that when looking at the legend, they thought that all genres were being shown. This was in fact not the case, since only a select few of the genres were being shown at the time the user was viewing the treemap.

- *Feedback:* In the treemap legend, include a cue that shows the user which genres are currently being viewed. This user also advised us to edit the description on the genres page to improve clarity.

Before having a 2nd user test our visualization, we made necessary changes facing
User 2:
- *Major/minor:* Public Health major, Astrophysics minor.
- *Notes on task completion:* This user also completed all the tasks successfully. When asked whether the description on the genre page was clear, they responded yes.
- *Feedback:* There was no feedback from this user.

User 3:
- *Major/minor:* Economics major, public health minor.
- *Notes on task completion:* This user was able to complete the tasks somewhat successfully. At one point during the task completion, the user made some wrong choices on the lowest selling games because they were solely relying on the size of the boxes. We also encouraged the user to use the tooltip to look at sales.
- *Feedback:* No feedback.

Overall, all our users were able to complete the four tasks effectively. We noticed that for some users, we needed to provide additional guidance on how to complete the tasks. This may be due to the fact that some of our tasks were not so intuitive or that our instructions were not so clear. For the last user tester, we noticed that when they were trying to observe the lowest selling game, they got it wrong because some of the games had similar box sizes. This can be considered a limitation of the treemap visualization because some of the games with small differences in sales did not have much variation in size. So, this can be somewhat confusing to a user who is trying to rank video games by how many sales were made. In spite of this, we are glad that the user can reference the tooltip to view the actual sales amounts and make correct comparisons.

Resources

Over the course of working on this project, we used the following resources:
- Dataset search: zenodo.org.
- ARIA documentation.
- Claude AI.
- Dr. William's office hours.
- D3 documentation (for treemap code).
- The WAVE help document provided by Dr. Williams.
- WAVE Chrome extension.
- This Data Viz Project website with examples of idioms → We drew some inspiration from this to get visualization ideas.

Personal Reflections

Asuna's reflection:

The overall collaboration process with Tania was very smooth and cooperative.

We started by meeting to find an interesting dataset, and we settled on a dataset that visualizes video game sales across different regions. We then brainstormed several visualization ideas together and decided to use treemaps to show sales grouped by different attributes. For the code implementation, we divided our work: Tania focused on implementing the genre page, while I focused on the treemap page. Later, we met several more times to work on the write-up and further refine the visualization based on feedback from students, pilot/user testing participants, and Dr. Williams.

Through this process, I learned more about the full experience of developing a visualization for a large dataset, from choosing the data and designing the visual structure to testing and improving the final product. I also learned how useful it is to seek feedback from others and observe how pilot testing participants actually interact with the visualization. Their feedback helped me notice many aspects of the design that I had overlooked at first, such as whether the visualization was intuitive for someone seeing it for the first time. This gave me a better understanding of what makes a visualization clear and easy to use.

I relied on Claude Code to assist with parts of my code implementation. More specifically, I first learned the D3 syntax for treemaps and built a working prototype of the treemap myself. Then, with the help of Claude Code, I added filters and made some style adjustments. I also used Claude to help combine the genre selection page and the treemap page into a more complete final visualization. Overall, this project helped me understand both the technical and design sides of data visualization more deeply.

Tania's reflection:

Overall, I enjoyed working on this project. First, I appreciated the level of detail that was included in the description of the project. Asuna and I were able to refer to it whenever we needed clarification during various stages of the project development.

Working with Asuna was very smooth and we communicated efficiently throughout the project duration. Together, we were able to come up with a visualization that we both were interested in implementing. One thing that really helped us was dividing tasks amongst each other, which we later brought together whenever we met outside class. I worked on developing the genre page while Asuna worked on the treemap page. Later on, she worked on merging the two pages together.

While working on the genre page (i.e. the default page), I relied on Claude AI to help me develop it. Some of the things that Claude helped with on this page include drawing the hexagons, creating a visually appealing interface, selection and deselection of genres, functionality of genre selection, explaining how to implement aria, creating live updates with aria, and adding specific functionalities to the keyboard navigation. I would say a big part of the code used in these areas was referenced from Claude.

Over the course of working on this project, I enjoyed learning more d3 syntax that I hope to continue using in future projects. I also learned that the process of developing a final working product involves a lot of iteration and refining. While developing the visualization, we would have new ideas, and talk about what we wanted to remove or improve, and it's great to see how that led us to having a functional visualization that we were both happy with. Additionally, another part of this implementation process that I enjoyed was working on accessibility features. In some of my previous classes, we've talked briefly about accessibility in various contexts. So, I appreciated having the opportunity to work on implementing accessibility features from a technological point of view, and hopefully, I get to work on implementing more complex accessibility features in the future.